IDCORE - Best Practice in the Supervision of Research Engineers

Updated May 2026

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Purpose of this document

This document has been created to provide clarity and guidance on the supervision of Engineering Doctorate (EngD) research projects conducted by Research Engineers on the IDCORE programme. The guide has been developed following a workshop held in Edinburgh on the 21st of February 2024. It should be read by all IDCORE Industrial and Academic supervisors and by our Research Engineers.

At the end of this guide is a supervision agreement. This shall be completed at the start of a research project, signed by the supervisors and the research engineer and sent to idcore.admin@ed.ac.uk. The agreement shall be updated as soon as practicable, if the supervisory team changes.

After the agreement is an appendix containing guidance for the Research Engineers on the project plan and annual progress reports that are submitted and assessed each year as part of the annual review process.

What is an EngD and how does it compare to a PhD?

Although at the same level academically, an EngD is quite different from a traditional PhD:

- The research is driven by the curiosity and commercial requirements of the industry stakeholder, rather than the curiosity of the research student and the academic supervisors.
- Usually, the EngD research is carried out at the company, and not on campus, so that the researcher is immersed in the industry experience and develops a sound understanding of the business implications of industrial research activity.
- Since the research is industry driven, there is a much greater tendency for EngD projects to be multidisciplinary, and create a more significant impact.
- Since the research is off campus, supervision is mostly remote.
- An EngD comprises 180 credits of taught material in addition to a 540-credit research project. In IDCORE 120 credits of the taught programme are delivered in year 1 and 60 credits are delivered during the research phase,
- An EngD viva differs from a traditional PhD viva in that a second External Examiner is appointed, from industry, to comment on the industrial impact of the research.

These fundamental differences mean that supervising an EngD researcher is very different to supervising a PhD student, on campus.

The IDCORE EngD

To cover the multi-disciplinary aspects of EngD projects, each IDCORE researcher has three academic supervisors, selected for their specialist skills from the partner institutions, as well as an industrial supervisor, and an Independent Reviewer. The EngD is a jointly awarded degree and the multi-institutional nature of the supervision teams is a requirement of the joint award.

Industrial supervision

All supervisors should be developing the person as well as the project.

The industrial company is expected:

- To provide a project supervisor who is able to have regular contact with the researcher, ideally regular face-to-face contact, at the premises of the company.
- To ensure continuity of supervision, if the allocated supervisor leaves the organisation, or if their role within the organisation changes, within a timeframe that allows an effective handover.

The industrial supervisor is expected:

- To devote time to supervising the project – around 1 hour per week in terms of contact time.
- To attend whole-supervision team meetings, as set out in the supervision agreement.
- To ensure that the researcher is fully onboarded with the company at the start of the project.

- To ensure that the researcher is integrated into, or aligned with, a project team.
- To ensure that the researcher has opportunities to be exposed to other commercial activities of the company.
- To ensure the researcher is fully briefed on, and able to comply with, all the company's Health & Safety procedures and policies.
- To fully brief an incoming supervisor, on IDCORE and the project status, in the event of a change in supervision.

On a discretionary basis, the researcher may spend up to 6 hours per week, on average, working on activities, for the company, that are not related to their own research project. Such activities are encouraged by the Centre, but must be at the discretion of the researcher, and must not distract the researcher from the main goal of completing their research project. The Centre expects that if unreasonable demands are placed on the researcher, the researcher will inform the Centre within a reasonable timeframe. Under such circumstances, the Centre will take corrective action.

Academic supervision

All supervisors should be developing the person as well as the project.

For each student, three supervisors will be appointed from the four partner institutions. The fourth institution will provide the 'Independent Reviewer' (see later).

The three IDCORE academic supervisors are **all equal** in terms of their supervision responsibilities. The 'lead' supervisor simply has the additional responsibility of dealing with the 'paperwork'. All supervisors have a duty of care to the student and should provide proactive pastoral support to the student when needed. This setup is different to the traditional PhD model, where there is one clear lead supervisor and one assistant supervisor. In recognition of this model the supervision effort recorded on the University of Edinburgh's student record system is split as 35% for the *lead-supervisor*, 31% for the two *co-supervisors*, with the remaining 3% allocated to the *independent reviewer.*

Each of the IDCORE supervisor's skills may not be fully required at all points during the project, nor even at all points during an individual meeting. But, even if specialist technical skills are not required from an individual supervisor, at a given point, that supervisor will have other valuable skills to offer, providing a broader view of the project as a whole, and therefore will often be very helpful in dealing with more generic things like structure, project management, dealing with difficulties, different approaches etc.

In summary: **all three** academic supervisors should **be fully engaged** at all points through the project.

Academic supervisors are expected:

- To do necessary university/institutional administration (e.g., fill in forms, select examiners)
- To teach knowledge and understanding (e.g., mini-tutorial in a topic area)
- To give technical advice in a domain (e.g., how to do an experiment, how to design something, what results mean)
- To help find a direction (e.g., setting of research questions)
- To give scholarly advice (e.g., how to write a paper, how to structure a paper, how to reply to a review)
- To help navigate the literature and state of the art (e.g., where to start, how to search)
- To give useful and timely feedback (e.g., on chapter drafts, on papers)
- To help navigate academic and professional networks (e.g., which conferences to attend, what to do there, how to work with an academic partner)
- To support project management (e.g., how to break down a big problem into stages, how to set objectives)
- To prepare for thesis submission and viva
- To encourage and to celebrate
- To provide pastoral support and to signpost to additional support when needed
- To mentor and give career advice

Supervision meetings

Roles and Responsibilities of each supervisor

At the start of the project, the newly gathered team needs to get to know each other (in-person – see below) and learn about their technical specialties, and when these skills may be most pertinent to the project.

All supervisors are responsible for non-technical aspects of the project – such as review of written work, project management, project planning, etc.

Each of the IDCORE supervisor's skills may not be fully required for the full duration of some meetings. As such it is often very helpful to cover the generic, broader aspects of a meeting first, with all supervisors, and then tackle the technical aspects later on in the meeting with a smaller group of specific supervisors. This can help with engagement, time management and streamlining of meetings, and researchers are encouraged to factor this sort of approach in to their meeting planning.

Meeting Frequency

Supervision meetings should be based on the requirements of the researcher. **The ideal meeting frequency, for the whole supervision team, is fortnightly**. However, at the very start of the project, weekly meetings might be required to guide the development of the project or during specific experimental phases, or during the thesis writing phase. However, some researchers may feel overly managed with a high frequency, and monthly meetings may be more appropriate.

Who should attend each meeting?

It is important that there are regular meetings with the *whole* supervision team. This helps the collaboration between industry and academia and helps ensure the project continues to meet both industry and academic requirements.

Supervision and guidance from the industrial supervisor should fit within company expectations and practices. Such meetings can be informal catchups, more formal one-to-one's, or part of regular team meetings (as appropriate). The Centre expects the industrial supervisor to be meeting at least once a week with the student.

A meeting schedule with the academic supervisors should be established at the beginning of the project. A weekly team meeting with the industrial supervisor joining fortnightly is a good option. The meeting schedule should be reviewed frequently to make sure that the researcher is receiving adequate support so that the project progresses successfully.

The first meetings – Project Kick-off

The IDCORE Symposium is held every year, mid-April, and this is an ideal time for supervisors to informally meet their new researchers in person – where possible. The Symposium includes talks from IDCORE researchers at various stages of their projects so it is an ideal way to learn about what IDCORE as a whole is achieving.

A formal Kick-off meeting should then be set up as close to the start of June as possible, in-person and with all supervisors present. It is a strong recommendation that this meeting happens at the company's premises. This will give the academic supervisors the opportunity to fully engage with the industrial aspects of the project, and to learn more about the activities of the sponsor company. Travel expenses for researchers and academic staff will be covered by IDCORE (Students and Edinburgh staff should book through Edinburgh's Key Travel and partner academics should book through their own travel partner and charge to their local IDCORE coding).

Guidance on coordinating meetings

It is the responsibility of the researcher to arrange supervision meetings, but with large teams, it can be very hard to coordinate. To aid coordination:

- Sharing calendars is encouraged so that the researcher can see viable slots for all invitees.
- Recurring meetings are encouraged as this is habit forming.
- Ensure that a recurring meeting continues to suit all, as other commitments come and go (eg teaching). Adjustments should be made as soon as a diary clash becomes apparent.

- Supervisors should always inform, asap, if they cannot attend a scheduled meeting – the researcher may wish to rearrange and advance notice is required for this.

In-person meetings

Remote supervision has always been a popular modality for IDCORE, due to the locations of all involved. Since the pandemic it has become a habitual, default way of working. It is generally agreed that in-person meetings are more productive, more engaging for participants, and more conducive to building relationships. As such, regular in-person meetings are strongly encouraged, and should be held at least twice per year. The IDCORE centre recognises that the level of commitment required to travel to in-person meetings is not insignificant, but nevertheless this is seen as a proportionate and fair request of supervisors given the benefits outlined above.

The locations of in-person meetings should rotate between the supervisor's home locations to make the travel commitment fair. Locations should include Exeter/Penryn, Swansea and Scotland. Occasionally it may be convenient to arrange meetings alongside conferences, project meetings and events where most of the supervisory team will already be. Travel expenses for researchers and academic staff will be covered by IDCORE. Travel expenses for researchers and academic staff will be covered by IDCORE (Students and Edinburgh staff should book through Edinburgh's Key Travel and partner academics should book through their own travel partner and charge to their local IDCORE coding).

If it's not working

Project directions can (and do) change after the initial proposal, when supervisors are allocated. If an academic supervisor really feels that they no longer have the skills to offer, and that the researcher would benefit from supervision by a colleague of theirs, then it is the supervisor's responsibility to suggest this to their institution's IDCORE Co-Director. They **must not** simply disengage from the project and stop supporting the researcher.

Similarly, if an academic supervisor's workload has changed and has become unmanageable since being appointed as supervisor, then it is the supervisor's responsibility to discuss this with their institution's IDCORE Co-Director. Once again, they **must not** simply disengage from the project and stop supporting the researcher.

In both cases, it is formally the responsibility of the IDCORE Co-Director from each institution to appoint a new supervisor.

Institutional Directors:

- Edinburgh: David Ingram (david.ingram@ed.ac.uk)
- Strathclyde: Julia Race (julia.race@strath.ac.uk)
- Exeter: Philipp Thies (P.R.Thies@exeter.ac.uk)
- Swansea: Ian Masters (I.Masters@Swansea.ac.uk)

If an industrial supervisor feels they no longer have the skills or time to properly support the researcher and the project, the supervisor is expected to discuss replacement options with their own line management.

Review and feedback of researcher output

Supervisors should be providing critical review of all their researcher's output, in a timely manner (*ideally* taking no longer than 3 weeks). Researchers should allow for sufficient time for supervisors to fit review and feedback into their workloads and should be cognisant of supervisors' fluctuating workloads and annual leave. Nevertheless, researchers are encouraged to include their expectations/explicit deadlines along with their feedback requests.

It's important to note that feedback need not be in written format, and verbal discussions at meetings are equally valuable.

Thesis review and feedback

At a reasonably early stage, ideally as part of the 3rd year Annual Review (AR), there should be a discussion on the structure and the overall narrative of the thesis. There should also be an agreed pattern for review and feedback of the thesis, that is workable for both researcher and supervisors. Thesis review is normally done chapter by chapter (or smaller elements). It is important to recognise that a detailed review on one chapter, covering formatting,

editorial and style aspects will apply to the whole document. It is also important to acknowledge that a thesis is a personal document, written by the researcher, with guidance from supervisors.

Pastoral support

In IDCORE supervision, all supervisors must take on an element of pastoral support, as they should all be meeting with the researcher regularly, and should get to know the researcher well enough to hopefully recognise signs of difficulty. However, within the structure of IDCORE there is additional and dedicated pastoral support from the Centre Administrator (Katrina Tait idcore.admin@ed.ac.uk), the Centre's Industrial Manager, (Madeleine Schia Madeleine.Schia@ei.ed.ac.uk) and from the Centre Director (David M Ingram David.Ingram@ed.ac.uk).

The IDCORE Centre wants to support researchers as much as possible – by helping in times of difficulty, and by being part of celebrations and achievements. In order to provide help and support, in times of difficulty, or for any reason, the Centre encourages all researchers, and staff, to start a conversation early, so that the best approach can be worked out, and so that an easily fixable problem does not progress to something more significant. Any information disclosed will be handled sensitively and shared only with those who need to know.

Quick links for signposting (but please always inform the Centre of any issues):

Edinburgh University's Student Wellbeing Service: https://student-wellbeing-service.ed.ac.uk/

Edinburgh University's Student Counselling Service: https://student-counselling.ed.ac.uk/

Independent Reviewer

The Independent Reviewer (IR) should be present at each Annual Review (AR) meeting to provide an independent view of the researcher's progress. After the AR meeting there should be an opportunity for the IR and the researcher to speak in private, giving the researcher an opportunity to voice any concerns about their supervision or their project. It is for this reason that it is very important that the IR is in fact independent of the supervisors, both professionally and socially.

The same IR should remain in place throughout the project, thus being a clear point of contact for the researcher at all points, and so that the IR can be aware of progress from the start, and any relevant history.

The IR and the researcher should meet at the start of the project – it is of benefit if this is part of the kick off meeting, but not critical. Thereafter the IR should attend each AR meeting.

Outside of the AR cycle, the IR should be somebody the researcher can approach for mentoring or support, if there are project or supervision difficulties.

The IR should be an experienced post-graduate research supervisor.

Student Travel to Scientific Conferences, Trade Shows and Hub Meetings

IDCORE researchers are provided with funds to attend ONE scientific conference during each year of their research project. They do not have to present a paper or poster during the first year of their project but **must present** after that. In addition, students are encouraged to attend trade shows and hub meetings (e.g. Supergen), such meetings do not count against the conference quota unless the student is presenting.

Supervisors must discuss appropriate scientific conferences and events with students and based on their experiences recommend conferences which will be **most beneficial**. Approval to attend should be sent by email to the student by every supervisor and then forwarded to the Centre Director and Administrator for formal approval. Supervisors are required to apply a considered approach to approval requests.

Students have a limited, fixed, budget for this kind of travel and must estimate the costs of the conference (using the excel workbook on the Wiki site) and the carbon emissions (in tons CO2e) associated with their trip. Best value must be sought at all times and students are to be encouraged to book early (using Key Travel), register early and

seek best value hotel accommodation. Any trade shows or hub meetings attended must also be covered by this budget.

Supervisors are expected to give approval (or a reason for refusing it) in a timely manner.

Access to Facilities

IDCORE researchers have access to large scale experimental facilities (FloWave, FastBlade, DMAC, Kelvin Hydrodynamics Lab, Swansea Wind Tunnel, etc) at the academic partners where this is needed for the project. Access to supercomputing facilities (including HPC-Wales) is also available. The need for access must be approved in advance by **all supervisors** and the proposed experimental programme discussed with the facility manager.

Industrial supervisors need to be aware that where the experiments generate significant benefit to the sponsor there is a risk of breeching state aid rules. To avoid this, we encourage cost sharing, with IDCORE covering the parts of the test which are largely for the benefit of the student, and the sponsor covering the parts of the test which are largely for their benefit.

Please remember that access needs to be booked well in advance and that there must be sufficient time left in the project for the results to be properly analysed. As with travel, approval for access should be sent by email to the student by every supervisor and then forwarded to the Centre Director and Administrator for formal approval.

Supervisors are expected to give approval for access (or a reason for refusing it) in a timely manner.

Consumables (including software and external training)

There is a modest budget for consumables, software licenses and external training available. Please check carefully if software licenses are available from any of the partner Universities, or the project sponsor, before seeking a permission to purchase software. Consumables must be properly costed, and University procurement processes followed. Please ensure that full quotes are obtained for any consumables, permission is given by every supervisor (by email, forwarded to the Centre's Director and Administrator). Please consider lead times on delivery and plan accordingly.

Supervisors are expected to give approval for procuring consumables (or their reason for refusing it) in a timely manner.

How to escalate difficulties

For both supervisors and researchers, it is important that there is a route to report or discuss any issues. In the first instance, and where appropriate, researchers and supervisors should feel comfortable to, respectfully, raise any issues with each other, at an early stage. All discussion should be professional and in-line with The University of Edinburgh's Dignity and Respect Policy.

If any party feels any matter may need escalation, the first port of call is to communicate with the relevant institutional director of IDCORE:

- Edinburgh: David Ingram (david.ingram@ed.ac.uk)
- Strathclyde: Julia Race (julia.race@strath.ac.uk)
- Exeter: Philipp Thies (P.R.Thies@exeter.ac.uk)
- Swansea: Ian Masters (I.Masters@Swansea.ac.uk)

However, the IDCORE Administrator is always on hand to offer support and advice and, with permission, can escalate any issue to the appropriate director.

IDCORE thesis guidance

As has already been stated the supervisory team should discuss the format of the thesis early in the 3rd year and

support the development of a thesis plan.

An EngD thesis differs from a traditional PhD thesis in that the novel contribution can also relate to business practice and procedures, and so it is expected that an EngD thesis will also consider the industrial impact of the work. Due to the industrial nature of an EngD, it is possible for an EngD thesis to follow a portfolio format, in which the thesis includes more than one area of investigation. It is important that a portfolio thesis details the philosophy connecting the different areas.

The supervisor team should guide the student to make an appropriate choice and discuss the structure of the document and its content.

Careful consideration should also be made of how confidential material is included in the thesis. The thesis is a public document and must be made freely available, and registered with the British Library, at some point. Experience has shown that where confidential material is necessary to the discussion it could be non-dimensionalised, or written so that a part of a chapter (or chapters) can be redacted. It is possible (with reason) to embargo the publication of the thesis for up to five years.

Under exceptional circumstances, it is possible to restrict access to a thesis a period longer than five years. It is important to realise that **no one**, not the candidate, their supervisors, the Head of School, the Sponsor Company or Library staff can have access to the thesis during this period. Whilst this option exists, careful planning should make it unnecessary. Should it be considered, it is important to remember that it is only allowed where the thesis contains:

- Intellectual property subject to a patent application,
- sensitive information whose disclosure would be an actionable breach of confidence or could prejudice the interests of an individual,

or where the signed sponsorship agreement with the sponsor or thesis partners includes restrictions on making the thesis publicly available.

IDCORE viva guidance

Prior to the viva

Approximately two months prior to the expected thesis submission, the researcher should complete Notice of Intention to Submit documents and submit these to the University of Edinburgh's College office (College SharePoint).

The 'lead' supervisor will then be called upon to nominate suitable examiners, by the Centre Administrator. The IDCORE Board of Directors will then assess the suitability of the nominated examiners. If approved, the nomination will be passed to the University of Edinburgh's College office. The 'lead' supervisor is then, normally, no longer involved in the viva process.

IDCORE EngD Examiners

An IDCORE EngD viva requires 3 examiners: an Internal Examiner (from any of the academic partners, not necessary from the institution of the lead supervisor), an External Academic Examiner and an Industrial Examiner (to comment on the industrial impact of the research).

Viva format and length

The viva format could be a brief presentation/overview by the researcher (entirely at the discretion of the Examiners) followed by questions by the Examiners, **wrapping up in about 3 hours**. It is strongly advised, by IDCORE, to not allow the viva to extend beyond 3 hours.

With two External Examiners there can be a tendency for both Externals to have their own separate agenda of questions for the candidate, thus making the viva last longer than is appropriate (for all parties, but especially the candidate).

The independent Internal Examiner can pose questions if they feel something hasn't been covered, but otherwise the primary role of the Internal Examiner is to ensure that the viva is fair and held according to regulations. The

Internal Examiner should ensure that questioning remains proportionate and should curtail lengthy discussions when necessary.

Questioning during the viva should cover all the points that require a formal correction – no correction should be a surprise to the candidate after the viva.

Organising the viva, including pre-meeting

It is the Internal Examiner's responsibility to organise the viva meeting itself. This includes setting the date and location for the viva, in discussion with all parties, including the student, discussing remote attendance requirements for any, or all, of the parties, organising the location and/or online platform for the viva.

While remote attendance for any, or all of the parties is permitted, in-person vivas are encouraged wherever possible.

The location of the viva can be held at any of the partner institutions, but examiners are encouraged to consider the wellbeing needs of the candidate, especially in terms of support available, after the viva.

It is strongly advised that a pre-viva meeting is also organised for all Examiners to attend. This meeting should facilitate streamlining of the questions from both External Examiners and an agreement of the corrections that are to be asked of the candidate.

If the pre-meeting suggests that a longer viva may be warranted, then this should be flagged to the IDCORE centre in advance.

If a particular thesis warrants longer discussion, then adequate breaks, including provision of refreshments and lunch should be imposed. Catering, charged to IDCORE, can be ordered in advance, and delivered to the examination venue.

The IDCORE Supervision Agreement

At the start of every project the researcher and the supervision team should discuss various elements of the expected supervision and all should sign the agreement (template below). This should be completed as soon after project kick off as possible. The signed document, as a scanned copy, or signed pdf, should be passed to the IDCORE Administrator by end of June.

This agreement should be reviewed on a regular basis to meet the changing requirements of the project as it progresses. The review frequency should be agreed upon at the outset, but at a minimum of twice per year, with the autumn Annual Review period one obvious juncture. Reviewed, signed, versions should be passed to the Administrator as per the agreed schedule.

Continued

IDCORE Supervision Agreement

This agreement seeks to formally record the various aspects required to achieve best practice in terms of IDCORE supervision, in order to best facilitate a successful project, completed on time and carried out by a well-supported IDCORE researcher.

Before completing this agreement, the researcher and all supervisors should consult the Best Practice Document for guidance. Please complete and sign the agreement as soon after project kick off as possible, and return a scanned copy or e-signed pdf, to the IDCORE Administrator by the end of June. Please submit reviewed versions to the Administrator as per the agreed schedule.

Research Engineer Name	
Project Title	
Date	

	Name	Institution/Company
Academic Supervisor		
Academic Supervisor		
Academic Supervisor		
Industry Supervisor		
Independent Reviewer		

We, the academic supervisors, agree:

- To attend meetings as set out below
- To attend in-person meetings as set out below
- To complete necessary university/institutional administration in a timely way
- To teach knowledge and understanding
- To give technical advice in a domain
- To help find a direction
- To give scholarly advice
- To help navigate the literature and state of the art
- To give useful and timely feedback on output
- To help navigate academic and professional networks
- To support project management
- To prepare for thesis submission and viva
- To encourage and to celebrate
- To commiserate and to signpost to support
- To mentor and give career advice
- To appropriately raise any issues regarding our abilities to continue to support the researcher and the project

I, the industrial supervisor, agree:

- To attend meetings as set out below
- To attend in-person meetings as set out below
- To attend whole-supervision team meetings, as set out below
- To ensure that the researcher is fully onboarded with the company at the start of the project
- To ensure that the researcher is integrated into, or aligned with, a project team
- To ensure that the researcher has opportunities to be exposed to other commercial activities of the company
- To ensure the researcher is fully briefed on, and able to comply with, all company Health & Safety procedures and policies
- To appropriately raise any issues regarding my ability to continue to support the researcher and the project

Continued

I, the IDCORE Research Engineer, agree:

- To coordinate meetings according to the schedule outlined below
- To be aware of the changing availability of all meeting attendees, and to adjust the schedule when necessary
- To submit output for feedback in a timely way, and with awareness of other commitments, including annual leave

The following meeting schedule will be used (please indicate which option has been agreed).

Agreed meeting frequency for academic and industry supervisor meetings		
Weekly	Fortnightly	Monthly

Agreed meeting frequency for academic-only meetings		
Weekly	Fortnightly	Monthly

Agreed meeting frequency for industry-only meetings		
Weekly	Fortnightly	Monthly

Agreed in-person meeting frequency for academic and industry supervisor meetings (at varying locations)		
Quarterly	Biannually	Annually

Review frequency of this Supervision Agreement (at Annual Review stage as minimum)		
Quarterly	Biannually	Annually

Next Review date

	Signed by	Date
Research Engineer		
Academic Supervisor		
Academic Supervisor		
Academic Supervisor		
Industry Supervisor		

IDCORE Annual Review Guidance – for researchers

End of Year 1 - Start of Research Phase – Project Planning

This process is worth 10 research credits. These credits must be awarded in order to progress to the 2nd year of the programme. All our PGR students are asked to go through this process in order to facilitate the early part of research projects – it should get you off to a reasonably focused start.
The Annual Review form opens up each year, automatically, in EUCLID, in the final 4 months of the academic year, 1st May – 31st Aug. You will receive automated notifications from EUCLID during the process.
The *IDCORE* process is as follows:

1. 1st May – online AR opens up on EUCLID - notifications sent to student and supervisors.
2. Diarise, **ASAP,** a Review Meeting with all your academic and industrial supervisors, including your Independent Reviewer (IR) (if known), for August. It is sensible to do this first/early on, so that you have a clear deadline to work towards. You will also need to contend with the difficulty of the availability of your supervisors. End of August is the deadline to have the whole process signed off.
3. Work on your document (requirements below) in discussion with all your supervisors. Use the attached Cover Sheet to keep track of your revisions. Finalise as much as possible.
4. Ask your industrial supervisor and IES HoRI (Head of Research Institute – Prof Alasdair McDonald) to sign off the cover sheet. Your Industrial Supervisor and HoRI have no further involvement in the online approval process.
5. Complete the student part of the online form on EUCLID and upload your Document (with Cover sheet attached).
6. Hold the Review Meeting so that all supervisors and IR can discuss and assess your project plan.
7. Complete any further revisions to your document and send final version to your supervisors and Katrina.
8. Your EUCLID online AR form will then pass between your supervisors in turn, so they can formally recommend your progression to the second year of the programme, followed by an IDCORE sign off and finally the Head of Graduate School sign off. Ideally this should all happen by 31st August (but in reality, online approvals take some time and can slip into September).
9. At the September Exam Board Meeting, the IDCORE Exam Board will endorse, or comment further on the recommendations of your supervisors.

What should the Submission Cover?

Your submission should include (in minimum 11pt font size):

- 3 pages only for a Literature Review
- 3 pages only for a Project Plan
- 1 page only for references
- 1 page only for a Gantt Chart (with year 1 in reasonable detail). **NB your plans must include an allocation of time for writing up your work – ideally at least 6 months.**
- Project Risk Assessment (5 key risks)
- Health & Safety Assessment, not restricted in length.
- RRI report, not restricted in length.

Please note that if the RRI or Project Plan indicate that there are ethical issues with your intended research, you will need to complete an additional Research Ethics and Integrity form – see cover sheet.

Useful resource: Research Toolkit - Scotland Beyond Net Zero. Although it is aimed primarily at providing impacts for funders, it may be of interest to help you quantify and identify the contribution made by your research projects. This can be helpful in project planning and in describing the impact of your work in the thesis/portfolio.

Continued

Your Document should address the following points (example sub headings):

What do you plan to do and why?

- What is your anticipated Contribution to Knowledge by the end of the project? Why is this research novel and important? What deliverables will there be from the project (such as pieces of kit, code or theory)?
- How will your work fit within the context of prior work?
- Give a clear statement of your Aims (what you hope to achieve by the end of the project).

How will you achieve this?

- What stages are likely to be needed to be necessary to complete your project?
- Give a precise list of measurable objectives (the broad steps you intend to follow to achieve your Aims).
- All research work (experiments, simulations, fieldwork, etc.) and thesis writing must be achievable within the three years. Include a diagrammatic work plan (Gantt chart, showing year 1 in reasonable detail) on a dedicated page.

What resources are required?

- What resources will you need for your research project? Resources might include lab space, equipment, materials, technical support time, data sets, software licenses, travel funds, training requirements, research costs, etc.
- Is your project reliant on collaboration with others? Does it rely upon either internal or external organisations to provide you with something?
- Are the necessary resources available; do they need to be purchased or arranged?

Risks to successful completion

- What are the risks to successfully completing your research work within the three years? How can you avoid delays due to these risks?
- Think particularly carefully about the resources and collaborations required for your project. If your project is critically reliant on things outside your control (e.g. parts being made, data being sent), think about whether your project plan is realistic. Are you being over-ambitious? How can you guarantee that things outside your control do not delay your work? Do you have an alternative plan?
- Unexpected results, changes in your project plans, waiting for data from external partners, or waiting for materials or equipment to become available are all part of a normal research project. These delays can all be reasonably anticipated through proper planning.

Notes

- You should write for a technical, but non-specialist reader.
- Prepare your report in consultation with your Supervisors, which might involve a few iterations and discussion. Allow plenty of time for your supervisors' input and feedback.
- ****REFERENCING****

Please be sure to be consistent with your referencing style – and maintain this throughout all your written work – all the way through to your thesis.

The recommendation is for all IDCORE students to use the **Harvard Style** – details here: https://www-citethemrightonline-com.ezproxy.is.ed.ac.uk/Basics/sample-text-and-reference-list-using-the-harvard-style

This is from the Cite Them Right/UoE website: https://www-citethemrightonline-com.ezproxy.is.ed.ac.uk/Home

End of Year 2 - End of Year Report

The work required for your 2nd year Annual Reviews is a progress report of what you have achieved so far, along with a plan of your future work. The process carries 10 research credits, to recognise the effort required.

On 1st May, your Online Annual Review process will open up for you as normal on EUCLID and should be completed/signed off by 31st August – ready for you to start your next year.

The *IDCORE* process is as follows:

1. From APRIL. Diarise, **ASAP**, a Review Meeting with all your academic and industrial supervisors, including your Independent Reviewer (IR) for ideally mid-August. It is sensible to do this first/early on, so that you have a clear deadline to work towards. You will also need to contend with the difficulty of the availability of your supervisors. End of August is the deadline to have the whole process signed off.
2. 1st May – online AR opens up on EUCLID – notifications sent to student and supervisors.
3. Work on your report (requirements below) in discussion with all your supervisors. Use the attached Cover Sheet to keep track of your revisions. Finalise as much as possible.
4. Ask your Industrial Supervisor to sign off your Cover Sheet (note this is a different Cover Sheet to 1st year). Your Industrial Supervisor will have no involvement in the online approval process.
5. Complete the student part of the online form on EUCLID and upload your Document (with Cover sheet).
6. Hold the Review Meeting so that all supervisors and IR can discuss and assess your submission and plans.
7. Complete any further revisions to your document and send final version to your supervisors and Katrina.
8. Your EUCLID online AR form will then pass between your supervisors in turn, so they can formally recommend your progression to the next year of the programme, followed by an IDCORE sign off and finally the Head of Graduate School sign off. Ideally this should all happen by 31st August (but in reality, online approvals take some time and can slip into September).
9. At the September Exam Board Meeting, the IDCORE Exam Board will endorse, or comment further on the recommendations of the review panel.

Your submission should include:

- Updated literature review (3 pages)
- Summary of progress to date (3 pages)
- Updated project plan (3 pages; focus on the next year)
- Impact Summary[1] (1 page; what impact will your research have?)
- References (1 page)
- Updated Gantt Chart (next 6 to 12 months in detail and **including an allocation of time for writing up your work – ideally at least 6 months)**
- Updated Risk Assessment and H&S Assessment
- Data/IP Security Assessment (1 page; this is instead of an updated RRI assessment and falls under 'Trusted Research' protocols – see here: https://www.npsa.gov.uk/specialised-guidance/trusted-research)
    1. A broad description of and reflection on your sponsor company's policies and procedures for securing and managing IP.
    2. Describe the actions you are taking in your project to protect IP from both being leaked to industrial competitors and from attacks and/or theft by hostile actors.
    3. Describe what protocols you would or will use if or when attending an international conference.

1. Useful resource: Research Toolkit - Scotland Beyond Net Zero. Although it is aimed primarily at providing impacts for funders, it may be of interest to help you quantify and identify the contribution made by your research projects. This can be helpful in project planning and in describing the impact of your work in the thesis/portfolio.

****REFERENCING****

Please be sure to be consistent with your referencing style – and maintain this throughout all your written work – all

the way through to your thesis.
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End of Year 3 - End of Year Report

The work required for your 3rd year Annual Reviews is a progress report of what you have achieved so far, along with a plan of your future work, and specifically a draft thesis plan. The process carries 10 research credits, to recognise the effort required.

On 1st May, your Online Annual Review process will open up for you as normal on EUCLID and should be completed/signed off by 31st August – ready for you to start your next year.

The *IDCORE* process is as follows:

10. From APRIL. Diarise, **ASAP**, a Review Meeting with all your academic and industrial supervisors, including your Independent Reviewer (IR) for ideally mid-August. It is sensible to do this first/early on, so that you have a clear deadline to work towards. You will also need to contend with the difficulty of the availability of your supervisors. End of August is the deadline to have the whole process signed off.
11. 1st May – online AR opens up on EUCLID – notifications sent to student and supervisors.
12. Work on your report (requirements below) in discussion with all your supervisors. Use the attached Cover Sheet to keep track of your revisions. Finalise as much as possible.
13. Ask your Industrial Supervisor to sign off your Cover Sheet (note this is a different Cover Sheet to 1st year). Your Industrial Supervisor will have no involvement in the online approval process.
14. Complete the student part of the online form on EUCLID and upload your Document (with Cover sheet).
15. Hold the Review Meeting so that all supervisors and IR can discuss and assess your submission and plans.
16. Complete any further revisions to your document and send final version to your supervisors and Katrina.
17. Your EUCLID online AR form will then pass between your supervisors in turn, so they can formally recommend your progression to the next year of the programme, followed by an IDCORE sign off and finally the Head of Graduate School sign off. Ideally this should all happen by 31st August (but in reality, online approvals take some time and can slip into September).
18. At the September Exam Board Meeting, the IDCORE Exam Board will endorse, or comment further on the recommendations of the review panel.

Your submission should include:

- Updated literature review (3 pages)
- Summary of progress to date (3 pages)
- Updated project plan (3 pages; focus on the next year)
- Impact Summary[1] (1 page; what impact will your research have?)
- References (1 page)
- Updated Gantt Chart (next 6 to 12 months in detail and **including an allocation of time for writing up your work – ideally at least 6 months).**
- Updated Risk Assessment
- Updated H&S Assessment
- Updated RRI/Ethics Assessment (updated from end of Year 1)
- Updated Data Security/IP assessment NOT required

1. Useful resource: Research Toolkit - Scotland Beyond Net Zero. Although it is aimed primarily at providing impacts for funders, it may be of interest to help you quantify and identify the contribution made by your research projects. This can be helpful in project planning and in describing the impact of your work in the thesis/portfolio.

****REFERENCING****

Please be sure to be consistent with your referencing style – and maintain this throughout all your written work – all the way through to your thesis.

The recommendation is for all IDCORE students to use the **Harvard Style** – details here:

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